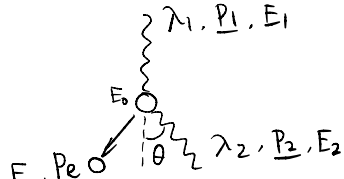
Compton Scattering

Before collision

$$E_0 = m_e c^2$$



After collision

$$E = \sqrt{m_e^2 c^4 + p_e^2 c^2}$$

$$= \sqrt{E_0^2 + p_e^2 c^2}$$

$$\therefore p_e^2 c^2 = E^2 - E_0^2$$

$$p_e = \frac{1}{c}\sqrt{E^2 - E_0^2}$$

Conservation of momentum

$$\underline{P_1} = \underline{P_2} + \underline{P_e}$$

where $P_1 = \frac{E_1}{c} = \frac{hf_1}{c} = \frac{h}{\lambda_1}$

$$P_2 = \frac{h}{\lambda_2}$$

$$\underline{P_e} = \underline{P_1} - \underline{P_2}$$

$$P_e^2 = P_1^2 + P_2^2 - 2P_1 \cdot P_2 \cos\theta \quad (1)$$

Conservation of energy

$$E_0 + E_1 = E + E_2$$

$$E_0 + p_1 c = \sqrt{E_0^2 + p_e^2 c^2} + p_2 c$$

$$\left[E_0 + (p_1 - p_2)c\right]^2 = E_0^2 + p_e^2 c^2$$

$$E_0^2 + (p_1 - p_2)^2 c^2 + 2cE_0(p_1 - p_2) = E_0^2 + p_e^2 c^2$$

$$p_e^2 = p_1^2 + p_2^2 - 2p_1 p_2 + 2\frac{E_0}{c}(p_1 - p_2) \qquad (2)$$

Combining (1) and (2) gives,

$$2p_1 p_2 \cos\theta = 2p_1 p_2 - 2\frac{E_0}{c}(p_1 - p_2)$$

$$\frac{E_0}{c}(p_1 - p_2) = p_1 p_2 (1 - \cos\theta)$$

$$\frac{p_1 - p_2}{p_1 p_2} = \frac{c}{E_0}(1 - \cos\theta)$$

$$\frac{1}{p_2} - \frac{1}{p_1} = \frac{1}{m_e c}(1 - \cos\theta)$$

$$\frac{\lambda_2}{h} - \frac{\lambda_1}{h} = \frac{1}{m_e c}(1 - \cos\theta)$$

$$\lambda_2 - \lambda_1 = \frac{h}{m_e c}(1 - \cos\theta)$$

$$\frac{c}{E_2} - \frac{c}{E_1} = \frac{1}{m_e c}(1 - \cos\theta)$$

$$\frac{1}{E_2} - \frac{1}{E_1} = \frac{1}{m_e c^2}(1 - \cos\theta)$$